Hogeschool van Amsterdam

SNMP lab

Version 2.0, 30-10-2019, by Ivo Nijhuis

Introduction

In this lab we'll get to know SNMP, the Simple Network Management Protocol. We'll set up two Virtual Machines (VM): one will play the part of the agent: a managed network device (such as a host, router, switch or any other device). The other VM will play the part of the managing server.

The agent will store information about its status in its Management Information Base (MIB) and the managing server can retrieve this information using SNMP.

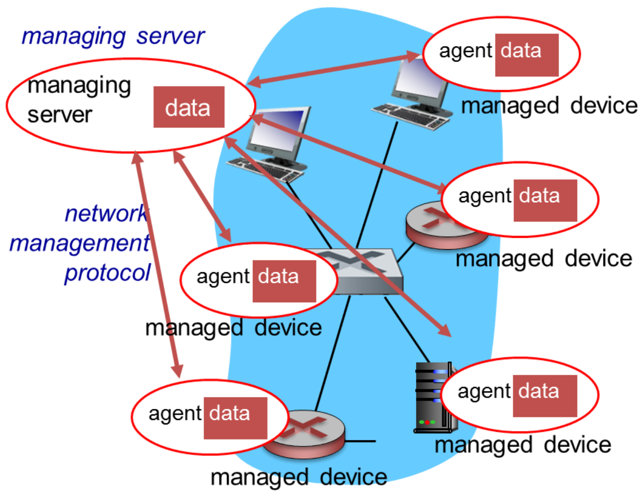

Figure 1, SNMP [Kurose & Ross]

Make sure to **read** Kurose and Ross Chapter 5.7 on SNMP before starting this lab!

Installation

Before you begin, you should have:

- VirtualBox installed on your laptop (You probably still have this, from when we set it up during the Infrastructure course. If not: download and install again.)
- Downloaded the Virtual Machine `HvA-NE1-Debian.ova` from the course materials on VLO

We want to set up two Virtual Machines that can communicate with each other. This means we'll need VirtualBox to set up a (virtual) private network on your host computer, so that the VM's each can get their own private IP address and can connect to each other.

The first step is to create such a private network in VirtualBox. In the menu, choose `File` and `Preferences`. Go to `Network` and add a new `NAT network`. Click `OK`.

The next step is to load the downloaded VM into VirtualBox. You may have already done this during the Nmap lab. If not, start VirtualBox and from the menu choose `File` and `Import Appliance`.

Once the import is complete, change the settings for this Virtual Machine. Under `General`, change the name into: `HvA-NE1-agent`.

Repeat the import or clone your existing VM, to create another VM. Change the name for this VM into `HvA-NE1-manager`.

Change the `Network` settings for both VMs. They must both be attached to the `NAT network` that you created earlier. Also, under `Advanced` network settings, make sure the two VMs have different `MAC addresses`. Change the MAC address for one of the VMs as needed.

Finally, you'll need the account information: you can login to your VMs using `student` / `HvA_student` or `root` / `HvA_root`.

Configuring the agent

Log in to your agent VM. Here, we'll install the SNMP daemon:

```
sudo apt-get update
sudo apt-get install snmpd
```

Now, we'll need to make a few changes to the configuration. With root access, open the file `/etc/snmp/snmpd.conf` in your favourite editor.

First, we need to change the agentAddress directive. Currently, it is set to only allow connections originating from the local computer. We need to comment out the current line, and uncomment the line underneath, which allows all connections.

```
#  Listen for connections from the local system only
#agentAddress  udp:127.0.0.1:161
#  Listen for connections on all interfaces (both IPv4 *and*
IPv6)
agentAddress udp:161,udp6:[::1]:161
```

We'll also need to add a user, that will be used from the manager to access the agent. Add the following lines, choosing your own username and password:

```
createUser username MD5 password
rwuser username
```

Save and close the file. Restart the snmpd service for your changes to take effect:

```
sudo service snmpd restart
```

Please note that the way we've configured things is not very secure in many ways. Please refer to [Digital Ocean] to find better ways of doing this. For now, we'll just leave our SNMP agent running in the background and start configuring the manager.

Configuring the manager

Now log in to your manager VM. Here, we'll install the SNMP manager, along with a package called `snmp-mibs-downloader`, which contains information about standard MIBs that allow us to access most of the MIB tree by name:

```
sudo apt-get update
sudo apt-get install snmp
sudo apt-get install snmp-mibs-downloader
```

The configuration is simple. With root access, open the file `/etc/snmp/snmp.conf` in your favourite editor and just comment out the one line:

```
#mibs :
```

Retrieving information from the MIB

Let's take our setup for a test drive. You'll need the username and password that you configured before in the agent. You will also need the agent's (private) IP address. (How can you find this?)

In the manager, enter the following command while inserting your own information:

```
snmpget -u username -l authNoPriv -a MD5 -A password
agent_ip_address 1.3.6.1.2.1.1.1.0
```

You should see a string describing the agent's system information, like this:

```
SNMPv2-MIB::sysDescr.0 = STRING: Linux debian 4.9.0-3-amd64 #1 SMP Debian 4.9.30
-2+deb9u5 (2017-09-19) x86_64
```

What just happened? We used the smpget command to retrieve a value from the agent's MIB. We added our authentication details to the command, as well as the agent's IP address. The final value, `1.3.6.1.2.1.1.1.0` , specified what object we wanted to retrieve from the agent's MIB.

Thanks to the `snmp-mibs-downloader` package we downloaded, we can also refer to objects by name. Try:

```
snmpget -u username -l authNoPriv -a MD5 -A password
agent_ip_address sysUpTime.0
```

How long has the agent been online?
Make a screenshot or picture of your results to hand in.

The Management Information Base

You can imagine the Management Information Base (MIB) as a tree of objects. Each object contains information about the agent that you can query.

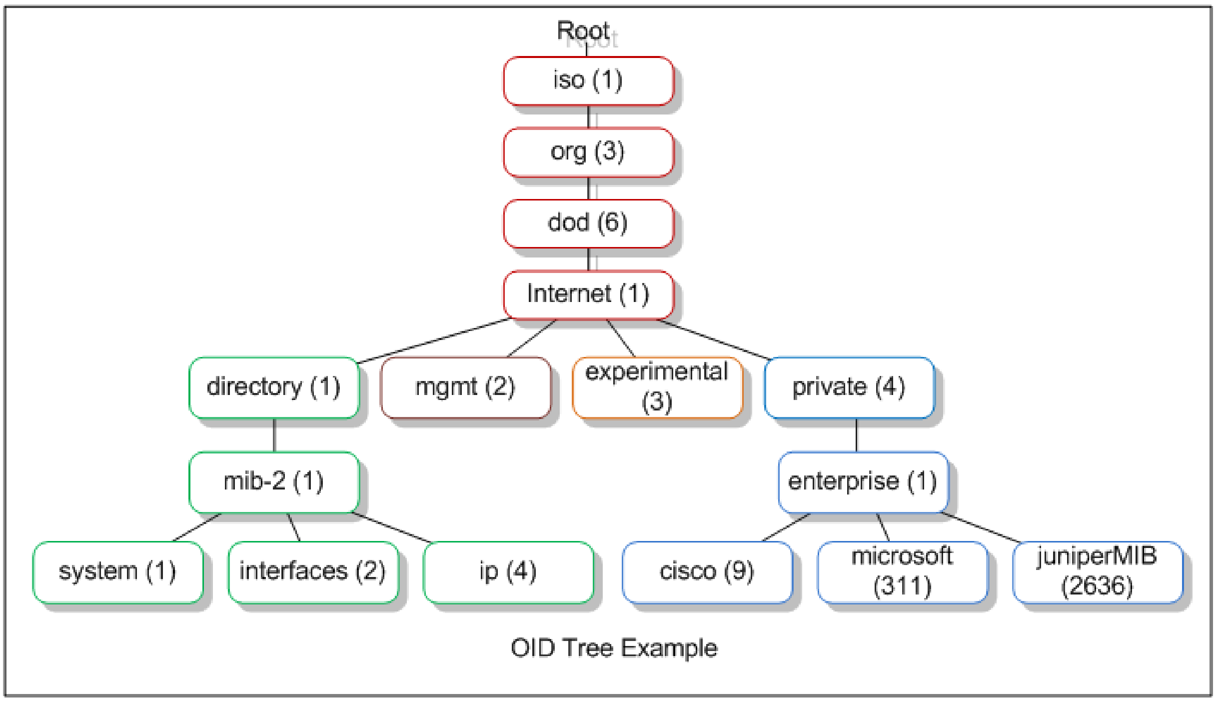

Figure 2, MIB tree [Leskiw]

Because of the structure of the tree, the objects we're interested in will usually start with 1.3.6.1.

Some more SNMP commands

First of all, repeating all of your authentication information for each SNMP command can get a bit tiring. Add the following lines to your `snmp.conf` file:

defSecurityName *username*
defSecurityLevel authNoPriv
defAuthType MD5
defAuthPassphrase *password*

Now, we'll no longer need to provide our authentication when using an SNMP command.

We can retrieve all values in a branch of the tree by using the `snmpwalk` command. Let's use this to retrieve all system information:

```
snmpwalk agent_ip_address system
```

Where is the agent located? (Hint: use `snmpwalk agent_ip_address system | more` to view the info per page.)

Now, use the *snmpwalk* command to query the *interfaces* branch of the tree to answer the questions below. Make a screenshot or picture of your results to hand in.

- What network interfaces are available on the agent? (You'll probably see two)
- How many packets were sent and received over each interface?
- Did any errors occur in the agent's network communication?

You could also monitor processes, diskspace or load on the agent by 'walking' the `prTable`, `dskTable` or `laTable` parts of the tree. The information in these trees is configured in the `snmpd.conf` file on the agent.

Finally, we can use the `snmpset` command to change values in the MIB. We can use this to change the location of the agent. First, we'll need to edit the `snmpd.conf` file on the agent and comment out the following lines:

```
#sysLocation  Sitting on the Dock of the Bay
#sysContact   Me <me@example.org>
```

Restart the `snmpd` service so that the location and contact will no longer be read-only values.

Now, use the `snmpset` command to set the agent location as you desire (the 's' indicates a string value):

```
snmpset agent_ip_address sysLocation.0 s "at the HvA"
```

Was your update succesful? Check this by using `snmpget` to retrieve the current value of `sysLocation.0` Make a screenshot or picture of your results to hand in.

Finally, use `smpset` to change the contact information into your own name and e-mailaddress.

Congratulations! You're well on your way to becoming an SNMP guru!

Sources

Please refer to the following sources to find more information:

- Digital Ocean, How To Install and Configure an SNMP Daemon and Client, retrieved on 26-10-2017 from https://www.digitalocean.com/community/tutorials/how-to-install-and-configure-an-snmp-daemon-and-client-on-ubuntu-14-04
- Kurose and K.W. Ross, 2016, Computer Networking, A Top-Down Approach, Pearson Education
- Leskiw, Aaron, SNMP tutorial part 2, retrieved on 26-10-2017 from http://www.networkmanagementsoftware.com/snmp-tutorial-part-2-rounding-out-the-basics/